

Balanced pancyclicity of the generalized base- b hypercube

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Abstract

Recently, Chan et al. have proposed a new cycle embedding property called *balanced pancyclicity* [Discrete Applied Mathematics 155(15) (2007) 1971-1978]. For a graph $G(V, E)$ and any two nodes x and y of V , a cycle R contains x and y can be divided into two paths, P_{t_1} and P_{t_2} , joining x and y such that $\text{len}(P_{t_1}) \leq \text{len}(P_{t_2})$, where $\text{len}(\lambda)$ denote the length of the path λ . A balanced cycle of an even (respectively, odd) length, $\text{len}(P_{t_1}) = \text{len}(P_{t_2})$ (respectively, $\text{len}(P_{t_1}) = \text{len}(P_{t_2}) - 1$). A graph is balanced pancyclic if every two nodes x and y are contained in a balanced cycle from $\text{Max}(3, 2\text{Dist}(x, y))$ to N , where N is the order of the graph. The interconnection network considered in this paper is the generalized base- b hypercube that is an attractive variant of the well-known hypercube. In fact, the generalized base- b hypercube is the Cartesian products of complete graphs with b nodes. The generalized base- b hypercube is superior to the hypercube in many criteria, such as diameter, connectivity and fault diameter. In this paper, we study balanced pancyclicity of the generalized base- b hypercube. We show that the generalized base- b hypercube is balanced pancyclic for $b \geq 4$.

1 Introduction

Recently, a new cycle-embedding property called balanced pancyclicity has been proposed [1]. Clearly, a cycle R contains x and y can be divided into two paths, P_{t_1} and P_{t_2} , joining x and y such that $\text{len}(P_{t_1}) \leq \text{len}(P_{t_2})$, where $\text{len}(\lambda)$ denote the length of the path λ ; that is denoted by $R = (P_{t_1} + P_{t_2})_{x,y}$. A balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ contains the P_{t_1} and P_{t_2} such that $\text{len}(P_{t_1}) = \text{len}(P_{t_2})$ for $\text{len}(R)$ is even and $\text{len}(P_{t_1}) = \text{len}(P_{t_2}) - 1$ for $\text{len}(R)$ is odd. A graph is balanced pancyclic if every two nodes x and y are contained in a balanced cycle from $\text{Max}(3, 2\text{Dist}(x, y))$ to N .

Many parallel computing architectures employ the bounded-degree interconnection networks as their topologies in the past decades. For examples, Tera Parallel Computer and CRAY T3D use the 2D torus and the 3D torus as their topologies, respectively [7]. The main drawbacks

with these interconnection networks are large average distances, large diameters and small bisection widths. Thus, they are not suitable for very high-performance computing [5, 10]. The n -dimensional generalized base- b hypercube, denoted by $GH(b, n)$, is superior to the above topological properties. However, $GH(b, n)$ has a serious scalable problem for traditional VLSI technology since each dimension of $GH(b, n)$ is a complete graph. No wonder that $GH(b, n)$ has not been extensively studied. Recently, to replace the traditional VLSI technology with the advanced technologies including the optical interconnection technology and the electronic switches, $GH(b, n)$ has been demonstrated to be a potential interconnection network for the next generation of parallel computing architectures [10].

$GH(b, n)$ has been proved to own many attractive properties such as *regularity*, *node transitivity* and *edge transitivity* [3, 5]. Informally, a node (respectively, edge) transitive graph looks the same when viewed from each node (respectively, edge). That a graph is node transitive or edge transitive implies that it is a symmetric graph to a high degree. Lakshmivarahan and Dhall shown that it possesses nice properties including lower diameter, Hamiltonicity, capability of embedding simple topologies [6]. Fragopoulou et al. investigated the primitive communication algorithms, such as one-to-all broadcasting, all-to-all broadcasting, one-to-all scattering and all-to-all scattering, on $GH(b, n)$ [3]. Wada et al. proposed a fault tolerant routing algorithm and discussed the fault-induced diameter on $GH(b, n)$ [9]. Ziavras and Krishnamurthy evaluated the broadcasting algorithms and the scattering algorithms on $GH(b, n)$ under realistic communication patterns [10]. Huang and Fang proved that $GH(b, n)$ is pancyclic, Hamiltonian-connected, panconnected and pancycle-connected for $b \geq 3$ [4, 5]. In this paper, we show that the generalized base- b hypercube is balanced pancyclic for $b \geq 4$.

2 Notations and background

A path of the length l is denoted by P_l ; and a cycle of the length s is denoted by C_s , where $s \geq 3$.

Definition 1. A Cluster-Torus(l, m) is $C_l \times K_m$.

Definition 2. A Cluster-Grid(l, m) is $P_{l-1} \times K_m$.

Definition 3. The n -dimensional generalized base- b hypercube, $GH(b, n)$, is defined recursively [6]:

1. $GH(b, 1)$ is K_b , where K_b is denoted for a complete graph with b nodes.
2. $GH(b, n)$ is $GH(b, n-1) \times K_b$ for $n \geq 2$.

That is, $GH(b, n)$ is $(K_b)^n$. Clearly $GH(b, n)$ is a generalization of the n -dimensional binary hypercube that is a $(K_2)^n$. $GH(b, n)$ comprises b^n nodes, each node x labeled by an n -digit number in radix b arithmetic $v_n v_{n-1} \dots v_2 v_1$. The node x is adjacent to another node y if and only if they differ by exactly one digit v_i , where $1 \leq i \leq n$ and (x, y) is called the dimensional i edge.

Proposition 1. The distance between two nodes x and y of $GH(b, n)$, denoted by $\text{Dist}(x, y)$, is the number of digit positions that x differs from y [6].

Definition 4. An automorphism of a graph $G = (V, E)$ is a permutation σ on V such that for each (x, y) in E if and only if $(\sigma(x), \sigma(y))$ is also in E [8].

Definition 5. A graph $G = (V, E)$ is node transitive if for each pair of nodes x and y in V , there exists an automorphism of G that maps x to y [8].

Definition 6. A graph $G = (V, E)$ is edge transitive if for each pair of edges (x_1, y_1) and (x_2, y_2) in E , there exists an automorphism of G that maps (x_1, y_1) to (x_2, y_2) [8].

Informally, a node (respectively, edge) transitive graph looks the same when viewed from each node (respectively, edge). That a graph possesses node transitivity or edge transitivity implies that it is a symmetric graph to a high degree.

Proposition 2. $GH(b, n)$ is both node transitive and edge transitive [3].

Clearly, the length of the smallest cycle containing two nodes x and y of a graph G must be greater than or equal to $2\text{Dist}(x, y)$. By definition, the length of a cycle must be greater than or equal to 3. Thus, we have:

Proposition 3. The length of the smallest cycle containing two nodes x and y of a graph G must be greater than or equal to $\text{Max}(3, 2\text{Dist}(x, y))$.

Proposition 4. $GH(b, n)$ is node-pancyclic and edge-pancyclic for $b \geq 3$ [4].

Proposition 5. $GH(b, n)$ is panconnected for $b \geq 3$ [5].

In this paper, we use $*$ as the unconcern symbol to label a subgraph of $GH(b, n)$. For example, $v_n v_{n-1} \dots v_2^*$ is the induced subgraph by the node set $S = \{v_n v_{n-1} \dots v_2 d \mid 0 \leq d \leq b-1\}$ [2]. Clearly, $v_n v_{n-1} \dots v_2^*$ is K_b . For example, the subgraph 00^* of $GH(3, 3)$, the induced subgraph of $GH(3, 3)$ by the node set $S = \{000, 001, 002\}$,

is K_3 . The outline graph of $GH(b, n)$, denoted by $\text{OG}(GH(b, n))$, is to take each $v_n v_{n-1} \dots v_2^*$ subgraph as a supernode, and the supernode $W^* = v_n v_{n-1} \dots v_2^*$ is adjacent to the supernode $U^* = u_n u_{n-1} \dots u_2^*$ in $\text{OG}(GH(b, n))$ if and only if $v_n v_{n-1} \dots v_2 d$ is adjacent to $u_n u_{n-1} \dots u_2 d$ for each $0 \leq d \leq b-1$ in $GH(b, n)$ [5]. Clearly, two supernodes $W^* = v_n v_{n-1} \dots v_2^*$ and $U^* = u_n u_{n-1} \dots u_2^*$ are adjacent if and only if they differ by exactly one digit position, the i th digit position, where $2 \leq i \leq n$. That is, the outline graph of $GH(b, n)$ is $GH(b, n-1)$. For example, as illustrated in Figure 2, the structure of $GH(3, 3)$ is shown. $\text{OG}(GH(3, 3))$ is to take each $v_3 v_2^*$ subgraph of $GH(3, 3)$ as a supernode, where $0 \leq v_3 \leq 2$, $0 \leq v_2 \leq 2$, and the supernode $v_3 v_2^*$ is adjacent to the supernode $u_3 u_2^*$ in $\text{OG}(GH(3, 3))$ if and only if $v_3 v_2 d$ is adjacent to $u_3 u_2 d$ for each $0 \leq d \leq 2$ in $GH(3, 3)$; that is, $v_3 v_2^*$ and $u_3 u_2^*$ differ by exactly one digit position. For instance, 01^* is adjacent to 00^* , 02^* , 11^* and 21^* . We have the following proposition:

Proposition 6. $\text{OG}(GH(b, n))$ is $GH(b, n-1)$ [4].

In a subgraph $W^* = v_n v_{n-1} \dots v_2^*$, the node $x = v_n v_{n-1} \dots v_2 d$ is said to be the node d of W^* . By the structure of $GH(b, n)$, the node d of W^* and the node d of U^* are adjacent if and only if W^* and U^* are adjacent in $\text{OG}(GH(b, n))$ for each $0 \leq d \leq b-1$. Clearly, if $\text{OG}(GH(b, n))$ embeds a cycle of supernodes with the length l , $(U_0^*, U_1^*, U_2^*, \dots, U_{l-1}^*, U_0^*)$, $GH(b, n)$ embeds the corresponding Cluster-Torus(l, b). Likewise, if $\text{OG}(GH(b, n))$ embeds a path of supernodes with the length l , $(U_0^*, U_1^*, U_2^*, \dots, U_{l-2}^*, U_{l-1}^*)$, $GH(b, n)$ embeds the corresponding Cluster-Grid(l, b). Thus, we state:

Proposition 7. If $\text{OG}(GH(b, n))$ embeds C_l , $GH(b, n)$ embeds Cluster-Torus(l, b). If $\text{OG}(GH(b, n))$ embeds P_{l-1} , $GH(b, n)$ embeds Cluster-Grid(l, b) [4].

Proposition 8. The lengths of the paths joining U_0 s_0 and $U_{l-1} s_1$ in the Cluster-Grid(l, b), $(U_0^*, U_1^*, U_2^*, \dots, U_{l-1}^*) \times K_b$, are in $\{l-1, l+1, l+2, \dots, bl-2, bl-1\}$ for $s_0 = s_1$ [5].

Proposition 9. The lengths of the paths joining U_0 s_0 and $U_{l-1} s_1$ in the Cluster-Grid(l, b), $(U_0^*, U_1^*, U_2^*, \dots, U_{l-1}^*) \times K_b$, range from l to $bl-1$ for $s_0 \neq s_1$ [5].

3 The results

In this section, we study the balanced pancyclicity of $GH(b, n)$. To investigate the balanced pancyclicity of $GH(b, n)$, the discussions about K_b , Cluster-Grid($2, b$) and Cluster-Torus(l, b) are required.

Lemma 1. For $b \geq 3$, given any two nodes $x = s_0$ and $y = s_1$ of K_b , there exists a balanced cycle $R =$

$(Pt_1+Pt_2)_{x,y}$ of each length ranging from 3 to b , where $s_0 \neq s_1$.

Proof. Without loss of generality, we assume that $x = 0$ and $y = 1$. There exists a path Pt_1 , $(0, b-1, b-2, \dots, t_1-1, t_1, 1)$, joining x and y of each length ranging from 2 to $b - \lceil b/2 \rceil = \lfloor b/2 \rfloor$, where $\lceil b/2 \rceil + 1 \leq t_1 \leq b-1$. There exists a path Pt_2 , $(1, 2, 3, \dots, t_2-1, t_2, 0)$, joining x and y of each length ranging from 2 to $\lceil b/2 \rceil$, where $2 \leq t_2 \leq \lceil b/2 \rceil$. Since they are edge-disjoint, there exists a balanced cycle $R = (Pt_1+Pt_2)_{x,y}$ of each length ranging from $2+2 = 4$ to $\lfloor b/2 \rfloor + \lceil b/2 \rceil = b$. Besides, there exists a balanced cycle of x and y , $(0, 1, 2, 0)$, with length 3. Thus, we know that there exists a balanced cycle $R = (Pt_1+Pt_2)_{x,y}$ of each length ranging from 3 to b . Q. E. D.

Lemma 2. For $b \geq 4$, given any two nodes $x = U_g s_0$ and $y = U_g s_1$ of Cluster-Grid(2, b), $P_1 \times K_b = (U_0^*, U_1^*) \times K_b$, where $0 \leq g \leq 1$, there exists a balanced cycle $R = (Pt_1+Pt_2)_{x,y}$ of each length ranging from 3 to $2b-1$, where $s_0 \neq s_1$.

Proof. Without loss of generality, we assume that $x = U_0 0$ and $y = U_0 1$. Since x and y locate in the same U_0^* subnetwork (i.e., K_b), from Lemma 1, there exists a balanced cycle $R = (Pt_1+Pt_2)_{x,y}$ of each length ranging from 3 to b . In the U_0^* subnetwork, there exists a path Pt_1 joining x and y of each length ranging from 1 to $b-1$. Likewise, in the U_1^* subnetwork, there exists a path λ joining $U_1 0$ and $U_1 1$ of each length ranging from 1 to $b-1$. Thus, there exists a path Pt_2 , $(U_0 0, \lambda, U_0 1)$, joining $U_0 0$ and $U_0 1$ of each length ranging from 3 to $b+1$. From the definition of a balanced cycle $R = (Pt_1+Pt_2)_{x,y}$, $\text{len}(Pt_1) = \text{len}(Pt_2)$ (respectively, $\text{len}(Pt_1) = \text{len}(Pt_2)-1$ for $\text{len}(R)$ is even (respectively, odd)). Hence, there exists a balanced cycle $R = (Pt_1+Pt_2)_{x,y}$ of each length ranging from $2+3 = 5$ to $b-1+b = 2b-1$. Combining the two results, there exists a balanced cycle $R = (Pt_1+Pt_2)_{x,y}$ of each length ranging from 3 to $2b-1$. Q. E. D.

Lemma 3. For $b \geq 4$, given any two nodes $x = U_g s_0$ and $y = U_g s_1$ of Cluster-Grid(2, b), $P_1 \times K_b = (U_0^*, U_1^*) \times K_b$, where $0 \leq g \leq 1$, there exists a balanced cycle $R = (Pt_1+Pt_2)_{x,y}$ of length $2b$, where $s_0 \neq s_1$.

Proof. Without loss of generality, we assume that $x = U_0 0$ and $y = U_0 1$.

Case 1: $b = 4$. Let $Pt_1 = (U_0 0 = x, U_1 0, U_1 2, U_0 2, U_0 1 = y)$, $Pt_2 = (U_0 1 = y, U_1 1, U_1 3, U_0 3, U_0 0 = x)$. Since they are edge-disjoint, there exists a balanced cycle $R = (Pt_1+Pt_2)_{x,y}$ of length $8 = 2b$.

Case 2: $b \geq 5$. We divide U_0^* into four subgraphs $G_{0,1}, G_{0,2}, G_{0,3}, G_{0,4}$ induced by $\{U_0 0\}$, $\{U_0 1, U_0 2, \dots, U_0 \lfloor b/2 \rfloor\}$, $\{U_0 \lfloor b/2 \rfloor + 1\}$, $\{U_0 \lfloor b/2 \rfloor + 2, U_0 \lfloor b/2 \rfloor + 3, \dots, U_0 b-1\}$, respectively. Likewise, we

divide U_1^* into two subgraphs $G_{1,1}$ and $G_{1,2}$ induced by $\{U_1 0, U_1 1, U_1 2, \dots, U_1 \lceil b/2 \rceil - 1\}$, $\{U_1 \lceil b/2 \rceil, U_1 \lceil b/2 \rceil + 1, \dots, U_1 b-1\}$, respectively. Clearly, $G_{0,2}, G_{0,4}, G_{1,1}$ and $G_{1,2}$ are all complete graphs. Let λ_1 be a Hamiltonian path of $G_{1,1}$ joining $U_1 0$ and $U_1 \lceil b/2 \rceil - 1$. Let λ_2 be a Hamiltonian path of $G_{0,2}$ joining $U_0 \lceil b/2 \rceil - 1$ and $U_0 1$. Let Pt_1 be $(U_0 0, \lambda_1, \lambda_2)$ whose length is $\lceil b/2 \rceil + \lfloor b/2 \rfloor = b$. Let λ_3 be a Hamiltonian path of $G_{1,2}$ joining $U_1 \lfloor b/2 \rfloor + 1$ and $U_1 \lfloor b/2 \rfloor + 2$. Let λ_4 be a Hamiltonian path of $G_{0,4}$ joining $U_0 \lfloor b/2 \rfloor + 2$ and $U_0 b-1$. Let Pt_2 be $(U_0 1, U_0 \lfloor b/2 \rfloor + 1, \lambda_3, \lambda_4, U_0 0)$ whose length is $(b-1 - \lceil b/2 \rceil) + (b-1 - (\lfloor b/2 \rfloor + 2)) + 4 = 2b - \lceil b/2 \rceil - \lfloor b/2 \rfloor = b$. Since they are edge-disjoint, there exists a balanced cycle $R = (Pt_1+Pt_2)_{x,y}$ of length $2b$. Q. E. D.

For example, let $b = 5$. $V(G_{0,1}) = \{U_0 0\}$, $V(G_{0,2}) = \{U_0 1, U_0 2\}$, $V(G_{0,3}) = \{U_0 3\}$, $V(G_{0,4}) = \{U_0 4\}$, $V(G_{1,1}) = \{U_1 0, U_1 1, U_1 2\}$, $V(G_{1,2}) = \{U_1 3, U_1 4\}$. Let $\lambda_1 = (U_1 0, U_1 1, U_1 2)$, $\lambda_2 = (U_0 2, U_0 1)$, $\lambda_3 = (U_1 3, U_1 4)$, $\lambda_4 = (U_0 4)$. We can derive $Pt_1 = (U_0 0, U_1 0, U_1 1, U_1 2, U_0 2, U_0 1)$ and $Pt_2 = (U_0 1, U_0 3, U_1 3, U_1 4, U_0 4, U_0 0)$. Since they are edge-disjoint, there exists a balanced cycle $R = (Pt_1+Pt_2)_{x,y}$ of length $10 = 2b$. From Lemma 2 and Lemma 3, we have:

Lemma 4. For $b \geq 4$, given any two nodes $x = U_g s_0$ and $y = U_g s_1$ of Cluster-Grid(2, b), $P_1 \times K_b = (U_0^*, U_1^*) \times K_b$, where $0 \leq g \leq 1$, there exists a balanced cycle $R = (Pt_1+Pt_2)_{x,y}$ of each length ranging from 3 to $2b$, where $s_0 \neq s_1$.

Similar to Lemma 4, we can derive:

Lemma 5. For $b \geq 4$, given any two nodes $x = U_0 s_0$ and $y = U_1 s_1$ of Cluster-Grid(2, b), $P_1 \times K_b = (U_0^*, U_1^*) \times K_b$, there exists a balanced cycle $R = (Pt_1+Pt_2)_{x,y}$ of each length ranging from 4 to $2b$, where $s_0 \neq s_1$.

Lemma 6. For any $b \geq 4$ and any odd $l = 2m+1 \geq 3$, given any two nodes $x = U_g s_0$ and $y = U_g s_1$ of Cluster-Torus(l, b), $C_l \times K_b = (U_0^*, U_1^*, U_2^*, \dots, U_{g-1}^*, U_g^*, U_{g+1}^*, \dots, U_{l-1}^*, U_0^*) \times K_b$, where $0 \leq g \leq l-1$ and $s_0 \neq s_1$, there exists a balanced cycle $R = (Pt_1+Pt_2)_{x,y}$ of each length ranging from $(l-1)b$ to lb .

Proof. Without loss of generality, we assume that $g = m$, $s_0 = 0$ and $s_1 = 1$.

Case 1: $l = 3$ (i.e., $m = 1$). Let G_1 be the subgraph induced by the node set $\{U_1 1, U_1 2, \dots, U_1 \lfloor b/2 \rfloor\}$. There exists a path λ_2 joining $U_2 0$ and $U_2 \lfloor b/2 \rfloor$ of each length ranging from 1 to $b-1$ in U_2^* . Let λ_1 be a Hamiltonian path of G_1 joining $U_1 \lfloor b/2 \rfloor$ and $U_1 1$. Let Pt_1 be $(U_1 0, \lambda_2, \lambda_1)$ that joins $U_1 0$ and $U_1 1$. The length of Pt_1 is ranging from $\lfloor b/2 \rfloor + 2$ to $\lfloor b/2 \rfloor + b$. Let G'_1 be the subgraph induced by the node set $\{U_1 \lfloor b/2 \rfloor + 1, U_1 \lfloor b/2 \rfloor + 2, \dots, U_1 b-1\}$. Let λ'_1 be a Hamiltonian path of G'_1 joining $U_1 \lfloor b/2 \rfloor + 1$ and $U_1 b-1$. There exists a path λ_0

joining $U_0 b-1$ and $U_0 0$ of each length ranging from 1 to $b-1$ in U_0^* . Let P_{t_2} be $(U_1 1, \lambda'_1, \lambda_0, U_1 0)$ that joins $U_1 1$ and $U_1 0$. The length of P_{t_2} is ranging from $(b-1-(\lfloor b/2 \rfloor + 1)) + 1 + 3 = b - \lfloor b/2 \rfloor + 2 = \lceil b/2 \rceil + 2$ to $(b-1-(\lfloor b/2 \rfloor + 1)) + (b-1) + 3 = 2b - \lfloor b/2 \rfloor = \lceil b/2 \rceil + b$. Since P_{t_1} and P_{t_2} are edge-disjoint, there exists a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ of each length ranging from $\lfloor b/2 \rfloor + 2 + \lceil b/2 \rceil + 2 = b + 4 \leq 2b$ to $\lfloor b/2 \rfloor + b + \lceil b/2 \rceil + b = 3b$ for $b \geq 4$.

Case 2: $l = 2m+1 \geq 5$. Let G_i be the subgraph induced by the node set $\{U_i 1, U_i 2, \dots, U_i b-1\}$ for each $m+1 \leq i \leq l-2$, let G_m be the subgraph induced by the node set $\{U_m 1, U_m 2, \dots, U_m \lfloor b/2 \rfloor\}$. Let path $\phi = (U_m 0, U_{m+1} 0, \dots, U_{l-3} 0, U_{l-2} 0)$. Clearly, there exists a path λ_{l-1} joining $U_{l-1} 0$ and $U_{l-1} b-1$ of each length ranging from 1 to $b-1$ in U_{l-1}^* . Let λ_i be a Hamiltonian path of G_i joining $U_i b-2$ and $U_i b-1$ for each $m+2 \leq i \leq l-2$:

$(U_i b-2, U_i b-3, U_i b-4, \dots, U_i 2, U_i 1, U_i b-1)$ if $i \equiv l-1 \pmod{2}$.

$(U_i b-1, U_i 1, U_i 2, \dots, U_i b-4, U_i b-3, U_i b-2)$ if $i \equiv l \pmod{2}$.

If $m+1 \equiv l-1 \pmod{2}$ (respectively, $m+1 \equiv l \pmod{2}$), let λ_{m+1} be a Hamiltonian path of G_{m+1} joining $U_{m+1} b-2$ (respectively, $U_{m+1} b-1$) and $U_{m+1} \lfloor b/2 \rfloor$. Let λ_m be a Hamiltonian path of G_m joining $U_m \lfloor b/2 \rfloor$ and $U_m 1$. Let P_{t_1} be $(\phi, \lambda_{l-1}, \lambda_{l-2}, \dots, \lambda_{m+1}, \lambda_m)$ that joins $U_m 0$ and $U_m 1$. The length of P_{t_1} is ranging from $(l-m-2)+2+(l-2-(m+1)+1)(b-1)+\lfloor b/2 \rfloor = mb-b+\lfloor b/2 \rfloor+2 = mb-\lceil b/2 \rceil+2$ to $(l-m-2)+b+(l-2-(m+1)+1)(b-1)+\lfloor b/2 \rfloor = mb+\lfloor b/2 \rfloor$.

Let G_i be the subgraph induced by the node set $\{U_i 1, U_i 2, \dots, U_i b-1\}$ for each $1 \leq i \leq m-1$, let G'_m be the subgraph induced by the node set $\{U_m \lfloor b/2 \rfloor + 1, U_m \lfloor b/2 \rfloor + 2, \dots, U_m b-1\}$. Let λ'_m be a Hamiltonian path of G'_m joining $U_m \lfloor b/2 \rfloor + 1$ and $U_m b-1$. Let λ_i be a Hamiltonian path of G_i joining $U_i b-2$ and $U_i b-1$ for each $1 \leq i \leq m-1$:

$(U_i b-2, U_i b-3, U_i b-4, \dots, U_i 2, U_i 1, U_i b-1)$ if $i \equiv m \pmod{2}$.

$(U_i b-1, U_i 1, U_i 2, \dots, U_i b-4, U_i b-3, U_i b-2)$ if $i \equiv m-1 \pmod{2}$.

If $0 \equiv m \pmod{2}$ (respectively, $0 \equiv m-1 \pmod{2}$), let d_0 be $b-2$ (respectively, $b-1$). Clearly, there exists a path λ_0 joining $U_0 d_0$ and $U_0 0$ of each length ranging from 1 to $b-1$ in U_0^* . Let path $\phi' = (U_1 0, \dots, U_{m-1} 0, U_m 0)$. Let P_{t_2} be $(U_m 1, \lambda'_m, \lambda_{m-1}, \dots, \lambda_1, \lambda_0, \phi')$ that joins $U_m 1$ and $U_m 0$. The length of P_{t_2} is ranging from $1+(b-1-(\lfloor b/2 \rfloor + 1)) + (m-1)(b-1) + 2 + m = mb - \lfloor b/2 \rfloor + 2$ to $1+(b-1-(\lfloor b/2 \rfloor + 1)) + (m-1)(b-1) + b + m = mb + b - \lfloor b/2 \rfloor = mb + \lceil b/2 \rceil$. Since P_{t_1} and P_{t_2} are edge-disjoint, there exists a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ of each length ranging from $mb - \lceil b/2 \rceil + 2 + mb - \lfloor b/2 \rfloor + 2 = 2mb - b + 4 \leq 2mb =$

$(l-1)b$ to $mb + \lfloor b/2 \rfloor + mb + \lceil b/2 \rceil = 2mb + b = lb$ for $b \geq$

4.

Q. E. D.

For example, let $l = 5$ (i.e., $m = 2$), $b = 4$, $x = U_2 0$ and $y = U_2 1$. That is, we consider Cluster-Torus(5, 4), $C_5 \times K_4 = (U_0^*, U_1^*, U_2^*, U_3^*, U_4^*, U_0^*) \times K_4$. Let G_2 be the subgraph induced by the node set $\{U_2 1, U_2 2\}$, $\phi = (U_2 0, U_3 0)$, λ_4 be a path joining $U_4 0$ and $U_4 3$ of each length ranging from 1 to 3 in U_4^* , $\lambda_3 = (U_3 3, U_3 1, U_3 2)$, $\lambda_2 = (U_2 2, U_2 1)$. $P_{t_1} = (\phi, \lambda_4, \lambda_3, \lambda_2)$ is a path joining $U_2 0$ and $U_2 1$ of each length ranging from 8 to 10. Let G'_2 be the subgraph induced by the node set $\{U_2 3\}$. Let λ'_2 be a Hamiltonian path of G'_2 joining $U_2 \lfloor b/2 \rfloor + 1 = U_2 3$ and $U_2 b-1 = U_2 3$. In this case, λ'_2 is degenerated to be the node $U_2 3$. Let $\lambda_1 = (U_1 3, U_1 1, U_1 2)$ and λ_0 be a path joining $U_0 2$ and $U_0 0$ of each length ranging from 1 to 3 in U_0^* , $\phi' = (U_1 0, U_2 0)$. $P_{t_2} = (U_2 1, \lambda'_2, \lambda_1, \lambda_0, \phi')$ is a path joining $U_2 1$ and $U_2 0$ of each length ranging from 8 to 10. Thus, there exists a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ of each length ranging from $8+8 = 16 = (l-1)b$ to $10+10 = 20 = lb$.

Lemma 7. For any $b \geq 4$ and any even $l = 2m \geq 4$, given any two nodes $x = U_g s_0$ and $y = U_g s_1$ of Cluster-Torus(l, b), $C_l \times K_b = (U_0^*, U_1^*, U_2^*, \dots, U_{g-1}^*, U_g^*, U_{g+1}^*, \dots, U_{l-1}^*, U_0^*) \times K_b$, where $0 \leq g \leq l-1$ and $s_0 \neq s_1$, there exists a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ of each length ranging from $(l-1)b+1$ to $lb-1$.

Proof. Without loss of generality, we assume that $g = m$, $s_0 = 0$ and $s_1 = 1$.

Case 1: $l = 4$ (i.e., $m = 2$). Let G_2 be the subgraph induced by the node set $\{U_2 1, U_2 2, \dots, U_2 b-1\}$. There exists a path λ_3 joining $U_3 0$ and $U_3 b-1$ of each length ranging from 1 to $b-1$ in U_3^* . Let λ_2 be a Hamiltonian path of G_2 joining $U_2 b-1$ and $U_2 1$. Let P_{t_1} be $(U_2 0, \lambda_3, \lambda_2)$ that joins $U_2 0$ and $U_2 1$. The length of P_{t_1} is ranging from $3+(b-2) = b+1$ to $2+(b-1)+(b-2) = 2b-1$. Let G_1 be the subgraph induced by the node set $\{U_1 1, U_1 2, \dots, U_1 b-1\}$. Let λ_1 be a Hamiltonian path of G_1 joining $U_1 1$ and $U_1 b-1$. There exists a path λ_0 joining $U_0 b-1$ and $U_0 0$ of each length ranging from 1 to $b-1$ in U_0^* . Let P_{t_2} be $(U_2 1, \lambda_1, \lambda_0, U_1 0, U_2 0)$ that joins $U_2 1$ and $U_2 0$. The length of P_{t_2} is ranging from $(b-2)+1+4 = b+3$ to $(b-2)+(b-1)+4 = 2b+1$. From the definition of a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$, $\text{len}(P_{t_1}) = \text{len}(P_{t_2})$ (respectively, $\text{len}(P_{t_1}) = \text{len}(P_{t_2}) - 1$ for $\text{len}(R)$ is even (respectively, odd)). Hence, there exists a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ of each length ranging from $(b+2)+(b+3) = 2b+5 \leq 3b+1$ (for $b \geq 4$) to $(2b-1)+2b = 4b-1 = lb-1$.

Case 2: $l = 2m \geq 6$. Let G_i be the subgraph induced by the node set $\{U_i 1, U_i 2, \dots, U_i b-1\}$ for each $m \leq i \leq l-2$. Let path $\phi = (U_m 0, U_{m+1} 0, \dots, U_{l-3} 0,$

$U_{l-2} 0$). Clearly, there exists a path λ_{l-1} joining $U_{l-1} 0$ and $U_{l-1} b-1$ of each length ranging from 1 to $b-1$ in U_{l-1}^* . Let λ_i be a Hamiltonian path of G_i joining $U_i b-2$ and $U_i b-1$ for each $m+1 \leq i \leq l-2$:

$(U_i b-2, U_i b-3, U_i b-4, \dots, U_i 2, U_i 1, U_i b-1)$ if $i \equiv l-1 \pmod{2}$.

$(U_i b-1, U_i 1, U_i 2, \dots, U_i b-4, U_i b-3, U_i b-2)$ if $i \equiv l \pmod{2}$.

If $m \equiv l-1 \pmod{2}$ (respectively, $m \equiv l \pmod{2}$), let λ_m be a Hamiltonian path of G_m joining $U_m b-2$ (respectively, $U_m b-1$) and $U_m 1$. Let P_{t_1} be $(\phi, \lambda_{l-1}, \lambda_{l-2}, \dots, \lambda_{m+1}, \lambda_m)$ that joins $U_m 0$ and $U_m 1$. The length of P_{t_1} is ranging from $(l-m-2)+2+(l-2-m+1)(b-1) = mb-b+1$ to $(l-m-2)+b+(l-2-m+1)(b-1) = mb-1$.

Let G_i be the subgraph induced by the node set $\{U_i 1, U_i 2, \dots, U_i b-1\}$ for each $1 \leq i \leq m-1$. Let λ_{m-1} be a Hamiltonian path of G_{m-1} joining $U_{m-1} 1$ and $U_{m-1} b-2$. Let λ_i be a Hamiltonian path of G_i joining $U_i b-2$ and $U_i b-1$ for each $1 \leq i \leq m-2$:

$(U_i b-2, U_i b-3, U_i b-4, \dots, U_i 2, U_i 1, U_i b-1)$ if $i \equiv m \pmod{2}$.

$(U_i b-1, U_i 1, U_i 2, \dots, U_i b-4, U_i b-3, U_i b-2)$ if $i \equiv m-1 \pmod{2}$.

If $0 \equiv m \pmod{2}$ (respectively, $0 \equiv m-1 \pmod{2}$), let d_0 be $b-2$ (respectively, $b-1$). Clearly, there exists a path λ_0 joining $U_0 d_0$ and $U_0 0$ of each length ranging from 1 to $b-1$ in U_0^* . Let path $\phi' = (U_1 0, \dots, U_{m-1} 0, U_m 0)$. Let P_{t_2} be $(U_{m-1}, \lambda_{m-1}, \dots, \lambda_1, \lambda_0, \phi')$ that joins $U_m 1$ and $U_m 0$. The length of P_{t_2} is ranging from $(m-1)(b-1)+2+m = mb-b+3$ to $(m-1)(b-1)+b+m = mb+1$. From the definition of a balanced cycle $R = (P_{t_1}+P_{t_2})_{x,y}$, $\text{len}(P_{t_1}) = \text{len}(P_{t_2})$ (respectively, $\text{len}(P_{t_1}) = \text{len}(P_{t_2})-1$) for $\text{len}(R)$ is even (respectively, odd). Hence, there exists a balanced cycle $R = (P_{t_1}+P_{t_2})_{x,y}$ of each length ranging from $(mb-b+2)+(mb-b+3) = 2mb-2b+5 \leq (l-1)b+1$ (for $b \geq 4$) to $mb-1+mb = 2mb-1 = lb-1$. Q. E. D.

For example, let $l = 6$ (i.e., $m = 3$), $b = 4$, $x = U_3 0$ and $y = U_3 1$. That is, we consider Cluster-Torus(6, 4), $C_6 \times K_4 = (U_0^*, U_1^*, U_2^*, U_3^*, U_4^*, U_5^*, U_0^*) \times K_4$. Let λ_5 be a path joining $U_5 0$ and $U_5 3$ of each length ranging from 1 to 3 in U_5^* , $\lambda_4 = (U_4 3, U_4 1, U_4 2)$, $\lambda_3 = (U_3 2, U_3 3, U_3 1)$, $\phi = (U_3 0, U_4 0)$. Let P_{t_1} be $(\phi, \lambda_5, \lambda_4, \lambda_3)$ that joins $U_3 0$ and $U_3 1$. The length of P_{t_1} is ranging from 9 to 11. Let λ_0 be a path joining $U_0 3$ and $U_0 0$ of each length ranging from 1 to 3 in U_0^* , $\lambda_2 = (U_2 1, U_2 3, U_2 2)$, $\lambda_1 = (U_1 2, U_1 1, U_1 3)$, $\phi' = (U_1 0, U_2 0, U_3 0)$. Let P_{t_2} be $(U_3 1, \lambda_2, \lambda_1, \lambda_0, \phi')$ that joins $U_3 1$ and $U_3 0$. The length of P_{t_2} is ranging from 11 to 13. Hence, there exists a balanced cycle $R = (P_{t_1}+P_{t_2})_{x,y}$ of each length ranging from $10+11 = 21 = (l-1)b+1$ to $11+12 = 23 = lb-1$.

Lemma 8. For any $b \geq 4$ and any even $l = 2m \geq 4$,

given any two nodes $x = U_g s_0$ and $y = U_g s_1$ of Cluster-Torus(l, b), $C_l \times K_b = (U_0^*, U_1^*, U_2^*, \dots, U_{g-1}^*, U_g^*, U_{g+1}^*, \dots, U_{l-1}^*, U_0^*) \times K_b$, where $0 \leq g \leq l-1$ and $s_0 \neq s_1$, there exists a balanced cycle $R = (P_{t_1}+P_{t_2})_{x,y}$ of length lb .

Proof. Without loss of generality, we assume that $g = 0$, $x = U_0 0$ and $y = U_0 1$.

Case 1: $b = 4$. Let $P_{t_1} = (U_0 0 = x, U_1 0, U_2 0, \dots, U_{l-2} 0, U_{l-1} 0, U_{l-1} 2, U_{l-2} 2, \dots, U_1 2, U_0 2, U_0 1 = y)$, $P_{t_2} = (U_0 1 = y, U_1 1, U_2 1, \dots, U_{l-2} 1, U_{l-1} 1, U_{l-1} 3, U_{l-2} 3, \dots, U_1 3, U_0 3, U_0 0 = x)$. Clearly, $\text{len}(P_{t_1}) = \text{len}(P_{t_2}) = 2l$. Since they are edge-disjoint, there exists a balanced cycle $R = (P_{t_1}+P_{t_2})_{x,y}$ of length $2l+2l = 4l = lb$.

Case 2: $b \geq 5$.

Subcase 2.1: m is odd. From Lemma 3, the case that $m = 1$ holds. Thus, in the following, we consider odd $m \geq 3$. Let G_i be the subgraph induced by the node set $\{U_i 0, U_i b-1, U_i b-2, \dots, U_i \lfloor b/2 \rfloor + 1\}$ for each $0 \leq i \leq m-1$, let G_i be the subgraph induced by the node set $\{U_i 0, U_i b-1, U_i b-2, \dots, U_i \lceil b/2 \rceil + 1\}$ for each $m \leq i \leq l-2$, and let G_i be the subgraph induced by the node set $\{U_i 1, U_i b-1, U_i b-2, \dots, U_i \lceil b/2 \rceil + 1\}$ for $i = l-1$. Let λ_i be a Hamiltonian path of G_i joining $U_i 0$ and $U_i b-1$ for each $0 \leq i \leq m-1$:

$(U_i 0, U_i \lfloor b/2 \rfloor + 1, U_i \lfloor b/2 \rfloor + 2, \dots, U_i b-2, U_i b-1)$ if i is even.

$(U_i b-1, U_i b-2, \dots, U_i \lfloor b/2 \rfloor + 2, U_i \lfloor b/2 \rfloor + 1, U_i 0)$ if i is odd.

Let λ_i be a Hamiltonian path of G_i joining $U_i 0$ and $U_i b-1$ for each $m \leq i \leq l-2$:

$(U_i 0, U_i \lceil b/2 \rceil + 1, U_i \lceil b/2 \rceil + 2, \dots, U_i b-2, U_i b-1)$ if i is even.

$(U_i b-1, U_i b-2, \dots, U_i \lceil b/2 \rceil + 2, U_i \lceil b/2 \rceil + 1, U_i 0)$ if i is odd.

Let λ_{l-1} be a Hamiltonian path of G_{l-1} joining $U_{l-1} b-1$ and $U_{l-1} 1$ since $l-1$ is odd. Let $P_{t_1} = (\lambda_0, \lambda_1, \dots, \lambda_{l-2}, \lambda_{l-1}, U_0 1)$. Clearly, $\text{len}(P_{t_1}) = m(b - \lfloor b/2 \rfloor) + m(b - \lceil b/2 \rceil) = 2mb - m(\lfloor b/2 \rfloor + \lceil b/2 \rceil) = mb$.

Let G'_i be the subgraph induced by the node set $\{U_i 1, U_i 2, U_i 3, \dots, U_i \lfloor b/2 \rfloor\}$ for each $0 \leq i \leq m-1$, let G'_i be the subgraph induced by the node set $\{U_i 1, U_i 2, U_i 3, \dots, U_i \lceil b/2 \rceil\}$ for each $m \leq i \leq l-2$, and let G'_i be the subgraph induced by the node set $\{U_i 0, U_i 2, U_i 3, \dots, U_i \lceil b/2 \rceil\}$ for $i = l-1$. Let λ'_i be a Hamiltonian path of G'_i joining $U_i 1$ and $U_i 2$ for each $0 \leq i \leq m-1$:

$(U_i 1, U_i \lfloor b/2 \rfloor, U_i \lfloor b/2 \rfloor - 1, \dots, U_i 3, U_i 2)$ if i is even.

$(U_i 2, U_i 3, \dots, U_i \lfloor b/2 \rfloor - 1, U_i \lfloor b/2 \rfloor, U_i 1)$ if i is odd.

Let λ'_i be a Hamiltonian path of G'_i joining $U_i 3$ and $U_i 2$ for each $m \leq i \leq l-2$:

$(U_i 3, U_i 4, \dots, U_i \lceil b/2 \rceil - 1, U_i \lceil b/2 \rceil, U_i 1, U_i 2)$ if i is even.

$(U_i 2, U_i 1, U_i \lceil b/2 \rceil, U_i \lceil b/2 \rceil - 1, \dots, U_i 4, U_i 3)$ if i is odd.

Let λ'_{l-1} be a Hamiltonian path of G'_{l-1} joining $U_{l-1} 2$ and $U_{l-1} 0$ since $l-1$ is odd. Let $P_{t_2} = (\lambda'_0, \lambda'_1, \dots, \lambda'_{l-2}, \lambda'_{l-1}, U_0 0)$ that joins $y = U_0 1$ and $x = U_0 0$. Clearly, $\text{len}(P_{t_2}) = m \lfloor b/2 \rfloor + m \lceil b/2 \rceil = m(\lfloor b/2 \rfloor + \lceil b/2 \rceil) = mb$. Hence, there exists a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ of length $mb + mb = 2mb = lb$.

Subcase 2.2: m is even. The proof is similar to the one of Subcase 2.1. Q.

E. D.

For example, let $l = 6$ (i.e., $m = 3$), $b = 5$, $x = U_0 0$ and $y = U_0 1$. Consider $\text{Cluster-Torus}(6, 5)$, $C_6 \times K_5 = (U_0^*, U_1^*, U_2^*, U_3^*, U_4^*, U_5^*, U_0^*) \times K_5$. Let $\lambda_0 = (U_0 0, U_0 3, U_0 4)$, $\lambda_1 = (U_1 4, U_1 3, U_1 0)$, $\lambda_2 = (U_2 0, U_2 3, U_2 4)$, $\lambda_3 = (U_3 4, U_3 0)$, $\lambda_4 = (U_4 0, U_4 4)$, $\lambda_5 = (U_5 4, U_5 1)$. Let $P_{t_1} = (\lambda_0, \lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5, U_0 1)$ that joins $x = U_0 0$ and $y = U_0 1$. The length of P_{t_1} is 15. Let $\lambda'_0 = (U_0 1, U_0 2)$, $\lambda'_1 = (U_1 2, U_1 1)$, $\lambda'_2 = (U_2 1, U_2 2)$, $\lambda'_3 = (U_3 2, U_3 1, U_3 3)$, $\lambda'_4 = (U_4 3, U_4 1, U_4 2)$, $\lambda'_5 = (U_5 2, U_5 3, U_5 0)$. Let $P_{t_2} = (\lambda'_0, \lambda'_1, \lambda'_2, \lambda'_3, \lambda'_4, \lambda'_5, U_0 0)$ that joins $y = U_0 1$ and $x = U_0 0$. The length of P_{t_2} is 15. Hence, there exists a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ of length $30 = 2mb = lb$. Combining Lemma 7 and Lemma 8, we have:

Lemma 9. For any $b \geq 4$ and any even $l = 2m \geq 4$, given any two nodes $x = U_g s_0$ and $y = U_g s_1$ of $\text{Cluster-Torus}(l, b)$, $C_l \times K_b = (U_0^*, U_1^*, U_2^*, \dots, U_{g-1}^*, U_g^*, U_{g+1}^*, \dots, U_{l-1}^*, U_0^*) \times K_b$, where $0 \leq g \leq l-1$ and $s_0 \neq s_1$, there exists a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ of each length ranging from $(l-1)b+1$ to lb .

Combining Lemma 6 and Lemma 9, we have:

Lemma 10. For any $b \geq 4$ and any $l \geq 3$, given any two nodes $x = U_g s_0$ and $y = U_g s_1$ of a $\text{Cluster-Torus}(l, b)$, $C_l \times K_b = (U_0^*, U_1^*, U_2^*, \dots, U_{g-1}^*, U_g^*, U_{g+1}^*, \dots, U_{l-1}^*, U_0^*) \times K_b$, where $0 \leq g \leq l-1$ and $s_0 \neq s_1$, there exists a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ of each length ranging from $(l-1)b+1$ to lb .

Lemma 11. For $b \geq 4$, given any two nodes x and y of $GH(b, n)$ such that $\text{Dist}(x, y) = 1$, there exists a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ of each length ranging from 3 to b^n .

Proof. We will prove the lemma by induction on n . For $n = 1$, $GH(b, 1)$ is K_b . From Lemma 1, the lemma holds.

Hypothesis: The lemma holds for some $n = k \geq 1$.

Induction Step: Consider two nodes x and y locate in $GH(b, k+1)$ and $\text{Dist}(x, y) = 1$. Without loss of generality, we assume that $x = u_{k+1} u_k \dots u_2 0$ and $y = u_{k+1} u_k \dots u_2 1$. Recall that $OG(GH(b, k+1))$ is

$GH(b, k)$. There exists an edge (U_0^*, U_1^*) in $OG(GH(b, k+1))$, where $U_0^* = u_{k+1} u_k \dots u_2^*$. Thus, the corresponding $\text{Cluster-Grid}(2, b)$, $(U_0^*, U_1^*) \times K_b$, is embedded in $GH(b, k+1)$. From Lemma 4, there exists a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ of each length ranging from 3 to $2b$. From Proposition 4, $OG(GH(b, k+1))$ is node-pancyclic. Thus, there exists a cycle, $(U_0^*, U_1^*, \dots, U_{l-1}^*, U_0^*)$, of $OG(GH(b, k+1))$, where $U_0^* = u_{k+1} u_k \dots u_2^*$ and $3 \leq l \leq b^k$. Clearly, there exists the corresponding $\text{Cluster-Torus}(l, b)$, $(U_0^*, U_1^*, \dots, U_{l-1}^*, U_0^*) \times K_b$, embedded in $GH(b, k+1)$. From Lemma 10, we know that there exists a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ of each length ranging in

$\{2b+1, 2b+2, \dots, 3b\}$ (for $l = 3$)

$\cup \{3b+1, 3b+2, \dots, 4b\}$ (for $l = 4$)

$\cup \{4b+1, 4b+2, \dots, 5b\}$ (for $l = 5$)

...

$\cup \{b^{k+1}-2b+1, b^{k+1}-2b+2, \dots, b^{k+1}-b\}$ (for $l = b^k-1$)

$\cup \{b^{k+1}-b+1, b^{k+1}-b+2, \dots, b^{k+1}\}$ (for $l = b^k$).

Thus, there exists a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ of each length ranging from $2b+1$ to b^{k+1} . Combining the results, there exists a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ of each length ranging from 3 to b^{k+1} . Q. E. D.

Lemma 12. For $b \geq 4$, given any two nodes x and y of $GH(b, n)$, if $OG(GH(b, n))$ embeds a balanced cycle of U_0^* and U_m^* with length l , $(U_0^*, U_1^*, U_2^*, \dots, U_{m-1}^*, U_m^*, U_{m+1}^*, \dots, U_{l-1}^*, U_0^*)$, where $x = U_0 s_0$, $y = U_m s_1$ and $s_0 \neq s_1$, there exists a balanced cycle $R = (P_{t_1} + P_{t_2})_{x,y}$ of each length ranging from $l+2$ to lb in $GH(b, n)$.

Proof. Consider the $\text{Cluster-Torus}(l, b)$, $(U_0^*, U_1^*, U_2^*, \dots, U_{m-1}^*, U_m^*, U_{m+1}^*, \dots, U_{l-1}^*, U_0^*) \times K_b$, embedded in $GH(b, n)$. Without loss of generality, let $x = U_0 0$ and $y = U_m 1$.

Case 1: l is odd. That is, $l = 2m+1$. Clearly the $\text{Cluster-Grid}(m, b)$, $(U_0^*, U_1^*, U_2^*, \dots, U_{m-1}^*) \times K_b$, is an induced subgraph of the $\text{Cluster-Torus}(l, b)$. From Proposition 8, there exists a path ϕ_1 joining $x = U_0 0$ and $U_{m-1} 0$ of each length ranging in $\{m-1, m+1, m+2, \dots, mb-1\}$ in the $\text{Cluster-Grid}(m, b)$. Let G_m be the subgraph induced by the node set $\{U_m 0, U_m b-1, U_m b-2, \dots, U_m \lceil b/2 \rceil + 1\}$. There exists a path λ_1 joining $U_m 0$ and $U_m b-1$ of each length ranging from 1 to $\lfloor b/2 \rfloor - 1$ in G_m . The length of $(\phi_1, \lambda_1, U_m 1)$ that joins $x = U_0 0$ and $y = U_m 1$ is ranging from $m+2$ to $mb + \lfloor b/2 \rfloor$. Besides, there exists a path η , $(U_0 0, U_1 0, \dots, U_{m-1} 0, U_m 0, U_m 1)$, of length $m+1$ joining x and y . Let P_{t_1} be a path in $\{(\phi_1, \lambda_1, U_m 1)\} \cup \{\eta\}$. The length of P_{t_1} is ranging from $m+1$ to $mb + \lfloor b/2 \rfloor$. Let G'_m be the subgraph induced by the node set $\{U_m 1, U_m 2, \dots, U_m \lceil b/2 \rceil\}$. There exists a path λ_2 joining $U_m 1$ and $U_m 2$ of each length ranging from 1 to $\lceil b/2 \rceil - 1$ in

G'_m . From Proposition 9, there exists a path ϕ_2 joining $U_{m+1} 2$ and $U_{l-1} 0$ of each length ranging from m to $mb-1$ in the Cluster-Grid(m, b), $(U_{m+1}^*, U_{m+2}^*, \dots, U_{l-1}^*) \times K_b$. The length of the path $(\lambda_2, \phi_2, U_0 0)$ that joins y and x is ranging from $m+3$ to $mb+\lceil b/2 \rceil$. Besides, there exists a path η' , $(U_m 1, U_{m+1} 1, \dots, U_{l-2} 1, U_{l-1} 1, U_{l-1} 0, U_0 0)$, of length $m+2$ joining y and x . Let P_{t_2} be a path in $\{(\lambda_2, \phi_2, U_0 0)\} \cup \{\eta'\}$. The length of P_{t_2} is ranging from $m+2$ to $mb+\lceil b/2 \rceil$. Since P_{t_1} and P_{t_2} are edge-disjoint, there exists a balanced cycle $R = (P_{t_1}+P_{t_2})_{x,y}$ of each length ranging from $(m+1)+(m+2) = 2m+3 = l+2$ to $mb+\lfloor b/2 \rfloor + mb+\lceil b/2 \rceil = 2mb+b = lb$.

Case 2: l is even. That is, $l = 2m$. The proof is similar to the one of Case 1. Q. E. D.

Theorem 13. For any $b \geq 4$, $GH(b, n)$ is balanced pancyclic.

Proof. We will prove the theorem by induction on n .

For $n = 1$, $GH(b, 1)$ is K_b . From Lemma 1, the theorem holds.

Hypothesis: The theorem holds for some $n = k \geq 1$.

Induction Step: Without loss of generality, consider two distinct nodes $u_{k+1} u_k \dots u_2 u_1$ and $v_{k+1} v_k \dots v_2 v_1$ in $GH(b, k+1)$, where $u_i \neq v_i$ for each $1 \leq i \leq \text{Dist}(u_{k+1} u_k \dots u_2 u_1, v_{k+1} v_k \dots v_2 v_1)$ and $u_i = v_i$ for each $1 + \text{Dist}(u_{k+1} u_k \dots u_2 u_1, v_{k+1} v_k \dots v_2 v_1) \leq i \leq k+1$.

Case 1: $\text{Dist}(u_{k+1} u_k \dots u_2 u_1, v_{k+1} v_k \dots v_2 v_1) \geq 3$ (i.e., $\text{Dist}(u_{k+1} u_k \dots u_2, v_{k+1} v_k \dots v_2) \geq 2$). Since $OG(GH(b, k+1))$ is $GH(b, k)$ and $GH(b, k)$ is balanced pancyclic, there exists a balanced cycle $R = (P_{t_1}+P_{t_2})_{x',y'}$ of each length ranging from $2\text{Dist}(u_{k+1} u_k \dots u_2, v_{k+1} v_k \dots v_2)$ to b^k in $OG(GH(b, k+1))$, where $x' = u_{k+1} u_k \dots u_2^*$ and $y' = v_{k+1} v_k \dots v_2^*$. From Lemma 12, there exists a balanced cycle $R = (P_{t_1}+P_{t_2})_{x,y}$ of each length ranging from $2\text{Dist}(u_{k+1} u_k \dots u_2, v_{k+1} v_k \dots v_2)+2 = 2\text{Dist}(u_{k+1} u_k \dots u_2 u_1, v_{k+1} v_k \dots v_2 v_1)$ to b^{k+1} in $GH(b, k+1)$.

Case 2: $\text{Dist}(u_{k+1} u_k \dots u_2 u_1, v_{k+1} v_k \dots v_2 v_1) = 2$ (i.e., $\text{Dist}(u_{k+1} u_k \dots u_2, v_{k+1} v_k \dots v_2) = 1$). Since $u_{k+1} u_k \dots u_2^*$ is adjacent to $v_{k+1} v_k \dots v_2^*$ in $OG(GH(b, k+1))$. The Cluster-Grid($2, b$), $(u_{k+1} u_k \dots u_2^*, v_{k+1} v_k \dots v_2^*) \times K_b$, is embedded in $GH(b, k+1)$. From Lemma 5, there exists a balanced cycle $R = (P_{t_1}+P_{t_2})_{x,y}$ of each length ranging from 4 to $2b$. From hypothesis, there exists a balanced cycle $R = (P_{t_1}+P_{t_2})_{x',y'}$ of each length ranging from 3 to b^k in $OG(GH(b, k+1))$, where $x' = u_{k+1} u_k \dots u_2^*$ and $y' = v_{k+1} v_k \dots v_2^*$. From Lemma 12, there exists a balanced cycle $R = (P_{t_1}+P_{t_2})_{x,y}$ of each length ranging from 5 to b^{k+1} in $GH(b, k+1)$. Combining the two results, we know that there exists a balanced cycle $R = (P_{t_1}+P_{t_2})_{x,y}$ of each length ranging from 4 to $2\text{Dist}(u_{k+1} u_k \dots u_2 u_1, v_{k+1} v_k \dots v_2 v_1)$ to b^{k+1} in $GH(b, k+1)$.

Case 3: $\text{Dist}(u_{k+1} u_k \dots u_2 u_1, v_{k+1} v_k \dots v_2 v_1) = 1$ (i.e., $\text{Dist}(u_{k+1} u_k \dots u_2, v_{k+1} v_k \dots v_2) = 0$). From Lemma 11, there exists a balanced cycle $R = (P_{t_1}+P_{t_2})_{x,y}$ of each length ranging from 3 to b^n . Q. E. D.

Consider two nodes x and y in $GH(3, n)$, where $n \geq 2$ and $\text{Dist}(x, y) = 1$. Without loss of generality, let $x = u_n \dots u_2 u_1$ and $y = u_n \dots u_2 v_1$. Since there exists only one path, $(u_n \dots u_2 u_1, u_n \dots u_2 w_1, u_n \dots u_2 v_1)$, of length 2 joining x and y , where u_1, v_1 and w_1 are three distinct digits, there is no balanced cycle $R = (P_{t_1}+P_{t_2})_{x,y}$ of length 4. We have:

Lemma 14. $GH(k, n)$ is not balanced pancyclic for $k = 3$ and $n \geq 2$.

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